

Tan Ngo

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EDUCATION

Hanoi University of Science and Technology (HUST)

Aug 2021 – Apr 2025

B.Sc. in Computer Science, Hanoi, Vietnam (GPA: 3.69/4.0)

- Graduated early in **3.5 years** and recognized as one of 4 outstanding graduating students (Director's Certificate of Merit).
- Selected to deliver the **graduation speech** on behalf of all graduating students.

Lam Son High School for the Gifted — *Specialized Mathematics Program, Vietnam*

Aug 2018 – May 2021

- **Vietnam National Mathematics Competition, Second Prize — TST List** (2021) — Ranked 11th nationally and selected for the Team Selection Test, the 2nd round for selecting Vietnam's International Mathematical Olympiad team.
- **Vietnam National Mathematics Competition, Consolation Prize** (2020) — National award in Vietnam's highest-level mathematics competition for gifted high school students.
- **North Central Region Mathematics Olympiad, First Prize** (2019) — Among specialized high schools in the North Central Region of Vietnam.

RESEARCH INTEREST

I began as a Generative AI Engineer focused on **parallel inference and optimization**, running experiments to find the fastest way to serve diffusion and language models in production. As an **AI Research Resident** on the Computer Vision Team at **Qualcomm AI Research Vietnam**, under the supervision of [Phong Ha Nguyen](#) (Senior Research Scientist, Qualcomm), [Binh-Son Hua](#) (Assistant Professor, Trinity College Dublin), and [Hung H. Bui](#) (VP, Qualcomm; ex-DeepMind), I worked on distilling **Image-Generation** models and building controllable **Video Diffusion Models, World Models**, including using them to generate training data for **Robot Learning**. Currently, I am passionate about **efficient ML**, **parallel and distributed computing**, and **ML systems**.

PUBLICATIONS

UniCaMo: Track the Noise, Move the World: 3D-Grounded Motion-Consistent Noise for Controllable Video Generation

- Long Van Vu*, **Tan Van Ngo***, Animesh Karnewar, Amir Habibian, Binh-Son Hua, Hung H. Bui, Minh Hoai, and Phong Ha Nguyen. *Under review at NeurIPS 2026*. · ***Co-first author** (equal contribution). · [Link](#)

RESEARCH AND WORK EXPERIENCE

AI Research Resident, Qualcomm AI Research Vietnam

Jun 2025 – Present

– UniCaMo - Unifying Camera and Motion Control for Video Generation (Jan 2026 – Present)

- Developing a method to jointly control camera movement and object motion in **3D space** in a video diffusion model, so a user can specify both the trajectory of the scene and the movement of subjects within it — a capability prior work typically handles only one of at a time.
- Introducing depth estimation as an explicit **3D signal** so the model reasons about scene geometry, rather than relying on **2D cues** alone, to improve motion consistency across frames.
- Scaling training, inference, and evaluation across **multi-node GPU clusters (up to 300 GPUs)** on RUNAI, working on data preparation, synchronization, throughput, and training stability.
- Fixed a model-loading default in **DiffSynth-Studio** that made all 96 GPUs (12 nodes × 8) queue behind one another at startup, re-downloading weights already on disk: per-node startup **9 min** → **~44 s (~12×)**, GPU start spread within a node **267 s** → **~11 s**, a recurring multi-node crash removed, inference throughput unchanged.

– Self-Directed Systems Research - Inspired from UniCaMo paper (2026 – Present)

- Favourite courses: [Stanford CS336](#) · [Karpathy's series](#) · [tinytorch](#) · [micrograd](#) · [GPU-MODE](#).
- Rebuilt my video-evaluation pipeline on **Ray** — GPU workers pulling from a shared queue plus a tree-reduction, replacing a shell launcher that split work unevenly and silently skipped data: **5.4× faster on 8 GPUs**.

- **Ray** wasn't slow — its defaults were, and fixing them sped up my own pipeline. The data transport ran at full hardware speed, but out of the box each worker got a single CPU thread and every step round-tripped through the driver — so CPU-bound work ran **4–9× slower** and coordination cost **~30× more time than computing**, with nothing crashing or warning.
 - A one-line config fix gave **6.7×** on the same hardware: on an 8× H100 node (198 CPU cores), one worker per GPU meant **Ray's** `num_cpus` acted as a scheduling token, not thread affinity — so CPU-bound metrics (LPIPS, SSIM, PSNR) ran single-threaded and serial. Setting `torch.set_num_threads(198//8)` to split the cores evenly across workers cut the stage **397.3 s → 59.7 s**.
 - Found that adding GPUs can make a pipeline slower: my evaluation ran fastest on **2 of 8 GPUs** and the depth/pose annotation on **4 of 8**, because the container's CPU budget is capped (198 cores) and is split among workers — past a point each extra GPU only waits for CPU.
- **Few-Step Text-to-Multiview Diffusion Model** (Jun 2025 – Dec 2025)
- **Distilled** a multi-step MVDream text-to-multiview model into a **few-step** student, sharply reducing the number of sampling steps while preserving multiview consistency, for faster interactive and on-device use.
 - Raised generation from 256×256 to 512×512 per view — 4× the pixels, above the resolution the teacher itself produced — and improved output quality through training-recipe changes, data refinement, and model optimization.
 - **Quantized** the few-step model and supported the applied team in deploying it on Samsung S24 phones; the result was shown as a live demo at **Qualcomm's NeurIPS 2025** booth.

Generative AI Engineer, ZenAI

Aug 2024 – Jun 2025

- **Diffusion Image-Generation Acceleration** (Aug 2024 – Jun 2025)
- **Speed up** image generation for FLUX and SDXL diffusion models in production:
 - **FLUX.1-schnell (8-step)**: context parallelism (2× H100) + `torch.compile` fusion + first-block caching cut inference **2.02 s → 0.86 s (≈2.35× faster)** at ≥95% image quality.
 - **SDXL Turbo (8-step)**: CFG parallelism (2× H100) + `torch.compile` cut inference **1.26 s → 0.72 s (≈1.75× faster)** at ≥99% image quality.
 - **Fixed a bug in** `torch.compile` that was silently blocking these speedups (torch 2.3.1): traced an opaque compiler error to a single operation whose index expression couldn't be built, then added a fallback so that operation degrades gracefully instead of killing the entire compilation — restoring the compiled path.
 - **Dynamic-shape problem** — found the limit of the technique itself: `torch.compile`'s speedup holds only for a small set of distinct input shapes (**~16**); beyond **~100 shapes** it silently falls back to uncompiled speed as each new shape triggers a recompilation. The fix: serve a small bucketed set of shapes rather than compiling per shape.
- **LLM Serving & Inference Optimization (vLLM, SGLang)** (Aug 2024 – Jun 2025)
- Served and optimized large language models with the vLLM and SGLang frameworks for production workloads.
 - Applied **quantization**, **KV-cache** optimization, and **parallel GPU** execution to cut **latency** and raise **throughput** under production serving loads.
 - Reduced **CPU–GPU** communication overhead to improve end-to-end serving **efficiency**.

Research Assistant, Computer Vision Lab, BKAI, HUST

Oct 2023 – Jan 2025

- **Continual Learning for Object Detection with Generative Data Replay** (Sep 2024 – Jan 2025)
- Applied continual learning to object detection to mitigate catastrophic forgetting as new tasks/classes arrive sequentially, instead of retraining on all data.
 - Used diffusion-generated replay data to stand in for past-task samples, improving knowledge retention and training stability without storing the original datasets.
- **Vision Transformers for Medical Endoscopy Image Segmentation** (Oct 2023 – Aug 2024)
- Researched Vision Transformer-based architectures for segmentation of medical endoscopy images, using Masked Autoencoder (MAE) self-supervised pretraining on unlabeled data to improve downstream tasks.

HONORS AND AWARDS

HUST Wings Scholarship — Awarded for 6 consecutive semesters.	2022–2024
HUST Academic Encouragement Scholarship — Computer Science department, <3% selected.	2021–2022

TECHNICAL SKILLS

Languages: Python, C/C++

Distributed & parallel: multi-GPU / multi-node training and inference (up to 300 GPUs), NCCL, CPU/GPU workload balancing, Ray, SLURM

Efficiency & compilers: `torch.compile`, quantization, KV-cache, distillation, vLLM, SGLang

Frameworks & libraries: PyTorch, Diffusers, ComfyUI, NumPy, OpenCV

REFERENCES

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